National University of Computer & Emerging Sciences, Lahore

AI2002 – Artificial Intelligence (BS-SE-5A) — Project Deliverable 6: Final Technical Report

How Far Does a Near-Perfect Speech Emotion Recogniser Generalise? Replication, Stress-Testing and Bias Analysis of a Lightweight Hand-Crafted-Feature Ensemble

Group ID	F26-11	**Track**	B – Research & Development
Members	22L-7786 (Lead), 23L-3002, 23L-3102	**Instructor**	Hajra Waheed
Repository	https://github.com/junaidshahid-dev/Ai-project — code, notebooks, trained models, demo app, README with setup and usage		

Abstract

Chowdhury et al. (*Scientific Reports*, 2025) report 97.57% accuracy on RAVDESS for a lightweight averaging ensemble of a 1D-CNN and a CNN-BiLSTM trained on four hand-crafted acoustic features (ZCR, RMS energy, MFCC, Chroma). We re-implemented the pipeline from the paper's specification and tested it under five evaluation protocols, against classical baselines, with a gender-bias check, a feature ablation and follow-up experiments that explain the results. With augmentation applied only to training clips, the ensemble reaches **81.9%**; applying augmentation *before* the train–test split, so that 99.1% of test samples have a near-copy in training, reproduces the paper's level (**96.5%**). For speakers never heard in training accuracy falls to **58.5% ± 1.9**, and on another corpus it is near chance (19.7% RAVDESS → TESS). All models are significantly less accurate for male than for female speakers (ensemble 50.7% vs 66.2%, $p < 0.001$). We show that the features identify the speaker (95.0%) better than the emotion (74.4%), that per-speaker normalisation raises unseen-speaker accuracy of an SVM from 49.3% to 59.2% without closing the gender gap, and that the paper's flattened input is sensitive to time shifts. The architecture is a sound lightweight model, but its headline accuracy is a property of the evaluation protocol.

Summary of the project

- **Goal:** check whether a published, near-perfect speech emotion recogniser still works on new clips, new speakers, a new dataset, and equally well for men and women (Track B: replication and analysis).
- **What we built:** the full pipeline (data indexing, feature extraction, augmentation, 5 models), 6 experiment protocols with 23 deep-model training runs, an evaluation and bias analysis, explanatory experiments, trained model files and a Streamlit demo app that predicts the emotion of an uploaded .wav file.
- **Main answer:** no. The reported accuracy depends on test clips leaking into training; for new speakers and new datasets the model is far weaker, and it is biased against male voices.

1. Introduction and Research Question

Speech emotion recognition (SER) classifies a speaker's emotional state from the voice alone, with uses in call-centre analytics, health screening and voice assistants. Lightweight models are attractive because they run on cheap hardware. The base paper's numbers (97.57% on RAVDESS, 100% on TESS) were obtained by training and testing on random splits of the same dataset, it does not say whether augmentation was applied before or after splitting, and the authors list cross-dataset validation as future work. **Research question:** *does the near-perfect accuracy of this lightweight model reflect emotion recognition that transfers to new clips, new speakers and new recording conditions, and is it equally accurate for male and female speakers?* We test six hypotheses:

	Hypothesis	Result
H1	Under the paper's protocol our implementation approaches the reported accuracy.	Not supported (81.9% vs 97.57%)
H2	Augmenting before splitting inflates accuracy.	Supported (96.5%)
H3	Accuracy drops substantially for unseen speakers.	Supported (58.5%)
H4	Accuracy collapses on an unseen dataset.	Supported (19.7% / 14.1%)
H5	Accuracy differs between male and female speakers.	Supported (–15.6 points for men)
H6	MFCC carries most information; Chroma adds a positive contribution.	Partly (MFCC yes, Chroma not confirmed)

2. Data

- **RAVDESS** (Kaggle): 1,440 speech clips from 24 professional actors (12 male, 12 female) speaking two fixed sentences in 8 emotions at normal and strong intensity, 48 kHz.
- **TESS** (Kaggle): 2,800 clips of "Say the word ___" with 200 words, spoken by two female actors (aged 26 and 64) in 7 emotions. Used as the unseen dataset.
- **Common labels.** As in the paper (288 neutral RAVDESS clips in its Table 1), RAVDESS *calm* is merged into *neutral* and TESS *pleasant surprise* becomes *surprised*: 7 classes (neutral, happy, sad, angry, fearful, disgust, surprised). RAVDESS is balanced except neutral (288 vs 192 clips); TESS has 400 per class. Duplicated files in the Kaggle copies are removed by file name.

3. Methodology and Architecture

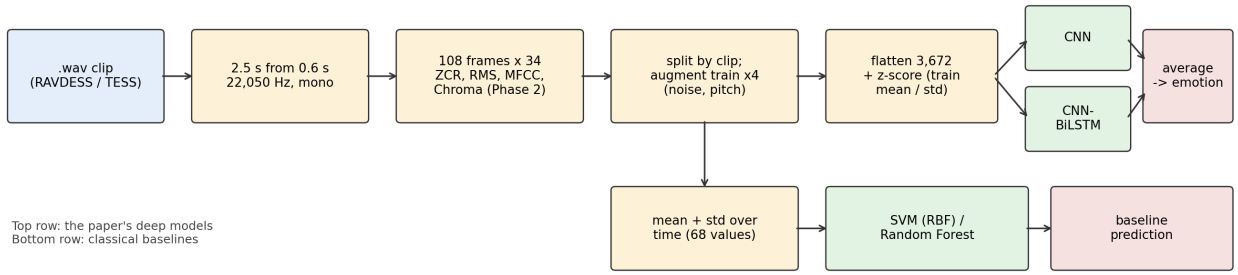


Figure 1. Pipeline from a .wav file to a predicted emotion. Top: the paper's deep models; bottom: our classical baselines.

Features and augmentation (paper Section 4). Audio is resampled to 22,050 Hz and 2.5 s are loaded from an offset of 0.6 s. With a 2,048-sample window and 512-sample hop this gives 108 frames; per frame we compute ZCR (1), RMS energy (1), 20 MFCCs and 12 Chroma values with librosa. Short clips are zero-filled. The 108 × 34 matrix is flattened to the paper's 3,672-value vector and z-scored with training-set statistics only. Each training clip is used in 4 versions: original, Gaussian noise, pitch shift (+0.7 semitones) and noise + pitch. Validation and test clips are never augmented (except in the deliberately leaky protocol).

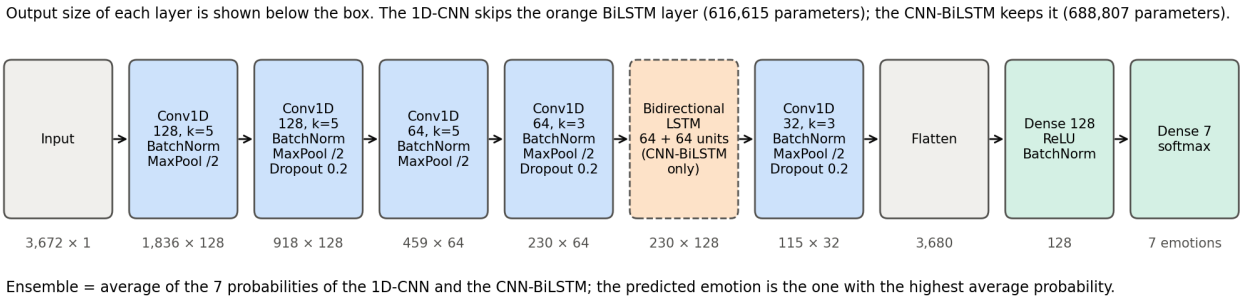


Figure 2. Architecture of the paper's 1D-CNN and CNN-BiLSTM (Table 2 of the paper) with the output size of every layer.

Models. (i) **1D-CNN**: five blocks of Conv1D → BatchNorm → MaxPool (128, 128, 64, 64, 32 filters), dropout 0.2 after blocks 2, 4 and 5, then Dense(128) and a 7-way softmax. (ii) **CNN-BiLSTM**: the same with a bidirectional LSTM (64 units per direction) before the last block. (iii) **Ensemble**: average of the two models' probabilities, the paper's final model. (iv) **Baselines**: majority class, an RBF-SVM (C = 10) and a 300-tree Random Forest on a 68-value summary (mean and std of each feature over time).

Training setting (paper Table 3)	Value
Optimiser, loss	Adam (learning rate 0.001), sparse categorical cross-entropy
Batch size, epochs	64, up to 100
Early stopping	on validation accuracy, patience 5, best weights restored
Learning-rate schedule	halved after 3 epochs without improvement (minimum 1e-5)
Seeds	42 for every split, model and library
Parameters	1D-CNN 616,615; CNN-BiLSTM 688,807 (paper: 672,423, see Section 10)

Evaluation protocols.

Protocol	Training data	Test data	Tests
P1 Random split (paper)	80% of RAVDESS clips + augmented copies; 10% validation	144 clips	H1
P2 Augment before split	80% of all 5,760 augmented samples	576 samples	H2
P3 Unseen speakers	4 folds; 18 actors train, 6 held out (3 M + 3 F)	all 1,440 clips once	H3, H5
P4 / P5 Cross-dataset	all RAVDESS (or all TESS)	all TESS (or RAVDESS)	H4
Ablation	P1 split, one feature group removed / MFCC only / no augmentation	144 clips	H6

Metrics: accuracy, macro-F1 (each emotion counts equally), weighted F1 (to compare with the paper), 95% bootstrap confidence intervals, McNemar's test between models, and a two-proportion z-test plus an actor-level Mann–Whitney test for the gender check.

4. Implementation Steps and Workflow

Python 3.13 with librosa 0.11 (audio and features), NumPy and pandas (arrays and tables), scikit-learn (baselines, splits, metrics), TensorFlow 2.21 / Keras 3 (deep models), SciPy (tests), Matplotlib (figures) and Streamlit (demo). All experiments run on CPU.

Step	Code	What happens
1. Index data	`src/ser/data.py`	find all .wav files, remove duplicates, map labels, read actor and gender
2. Features	`scripts/extract_features.py`	4 versions × 108 frames × 34 features per clip, saved once to features/*.npz
3. Models	`src/ser/models.py`	baselines and the paper's 1D-CNN / CNN-BiLSTM / ensemble
4. Experiments	`scripts/run_experiments.py`	trains every model under every protocol, saves per-clip predictions in results/
5. Evaluation	`phase4/F26-11.ipynb`	metrics, confidence intervals, model comparison, errors, gender bias
6. Analysis	`phase5/F26-11.ipynb`	reproducibility check and follow-up experiments
7. Use the model	`app/app.py`	upload a .wav file → features → normalise → both models → average → emotion

The deliverable notebooks follow the same steps in a simpler, self-contained form: Phase 2 (data and features), Phase 3 (training the four models and predicting a raw file), Phase 4 (evaluation) and Phase 5 (analysis). The final models used by the app are trained on all of RAVDESS and saved as `models/CNN.keras`, `models/CNN_BiLSTM.keras` and `models/normalizer.npz` (training mean and std).

5. Results and Evaluation Summary

Model	P1 Random split	P2 Augment first	P3 Unseen speakers	P4 RAV → TESS	P5 TESS → RAV
Majority (chance)	20.1 [14–27] F1 4.8	20.0 [16–24] F1 4.8	20.0 [18–22] F1 4.8	14.3 [13–16] F1 3.6	20.0 [18–22] F1 4.8
SVM	77.1 [70–83] F1 76.7	85.6 [83–89] F1 85.4	49.1 [47–52] F1 48.8	23.9 [22–26] F1 17.2	13.6 [12–15] F1 7.7
Random Forest	69.4 [62–76] F1 68.0	71.5 [68–75] F1 70.5	46.8 [44–49] F1 43.3	31.1 [29–33] F1 23.8	14.4 [13–16] F1 5.6
1D-CNN	79.2 [72–85] F1 78.6	95.7 [94–97] F1 95.7	56.7 [54–59] F1 56.0	17.8 [16–19] F1 14.3	14.1 [12–16] F1 6.6
CNN-BiLSTM	81.2 [75–88] F1 80.6	94.4 [92–96] F1 94.5	56.5 [54–59] F1 56.2	17.9 [16–19] F1 17.1	13.6 [12–15] F1 4.0
Ensemble	81.9 [76–88] F1 81.4	96.5 [95–98] F1 96.6	58.5 [56–61] F1 57.8	19.7 [18–21] F1 17.0	14.1 [12–16] F1 6.6

Table 1. Test accuracy % [95% confidence interval] and macro-F1 %. Chance (majority class) is 20.0% on RAVDESS and 14.3% on TESS.

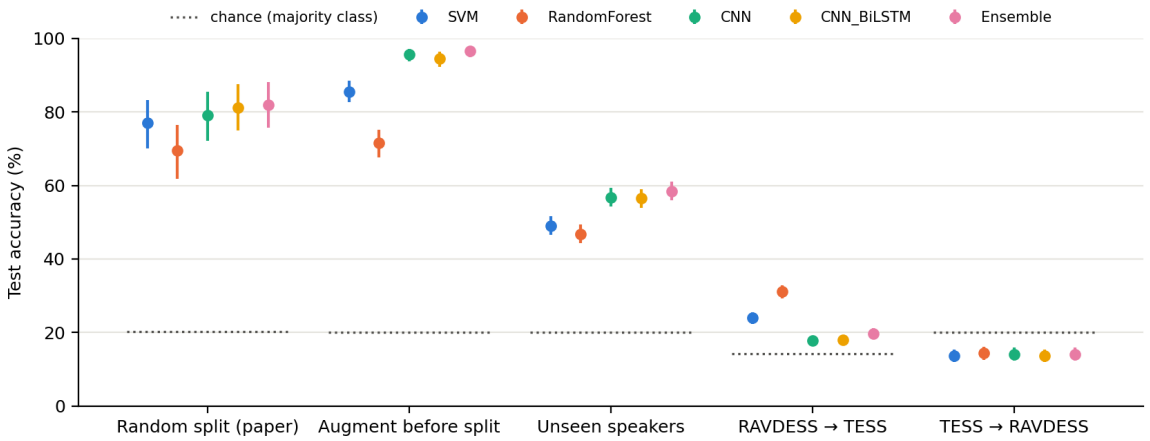


Figure 3. Accuracy of every model under every protocol with 95% confidence intervals.

- **The protocol matters more than the model:** the same ensemble ranges from 96.5% (P2) to 81.9% (P1), 58.5% (P3) and near chance (P4, P5).
- **Deep models help within a corpus:** for unseen speakers the ensemble beats the SVM by 9.4 points (McNemar $p < 0.001$). On the 144-clip random split the 4.9-point lead is not significant ($p = 0.30$).
- **Ensembling helps a little:** +1.8 points over the 1D-CNN for unseen speakers ($p = 0.04$); the CNN-BiLSTM is not better than the 1D-CNN.
- **Efficiency:** on a 4-core CPU the 1D-CNN trained in 10.6 min (21 epochs) and the CNN-BiLSTM in 13.1 min (23 epochs); prediction takes 2.7 and 4.1 ms per clip, the SVM 0.3 ms. On our 2-core laptop one epoch takes about 3.7 min.

6. Replication Results vs Reported Paper Results

Model (RAVDESS)	Reported in paper: acc / weighted F1	Our replication, clean split (P1)	Our replication, augment before split (P2)
1D-CNN	96.18 / 96.22	79.17 / 79.04	95.66 / 95.65
CNN-BiLSTM	97.57 / 97.29	81.25 / 81.14	94.44 / 94.44
Ensemble	97.57 / 97.56	81.94 / 81.94	96.53 / 96.53
Ensemble on TESS	100.00 / 100.00	not tested within TESS	–

Table 2. Accuracy and weighted F1 (%) reported by Chowdhury et al. (Table 4) and obtained by us.

- With a leak-free split our implementation is **15.6 points below** the reported ensemble accuracy. The order of the models matches the paper (ensemble ≥ CNN-BiLSTM ≥ 1D-CNN).
- Augmenting before splitting reproduces the paper's numbers within about one point (96.5% vs 97.57%; 1D-CNN 95.7% vs 96.18%). In that setting 99.1% of test samples have a noisy or pitch-shifted copy of the same recording in training, and accuracy is 97.8% on original clips and 95.6% on noise + pitch copies.
- The paper does not state its split order, so we cannot prove that this is what happened, but it is the only one of our settings that reproduces its results. The inflation is much larger for the deep models (+14.6 points for the ensemble) than for the SVM (+8.5), consistent with memorising near-duplicates.
- Our CNN-BiLSTM has 688,807 parameters instead of 672,423: the paper's Table 2 lists 6,176 parameters for the last convolution, which implies 64 input channels although the Bi-LSTM before it outputs 128. We followed the layer description.

7. Error Analysis

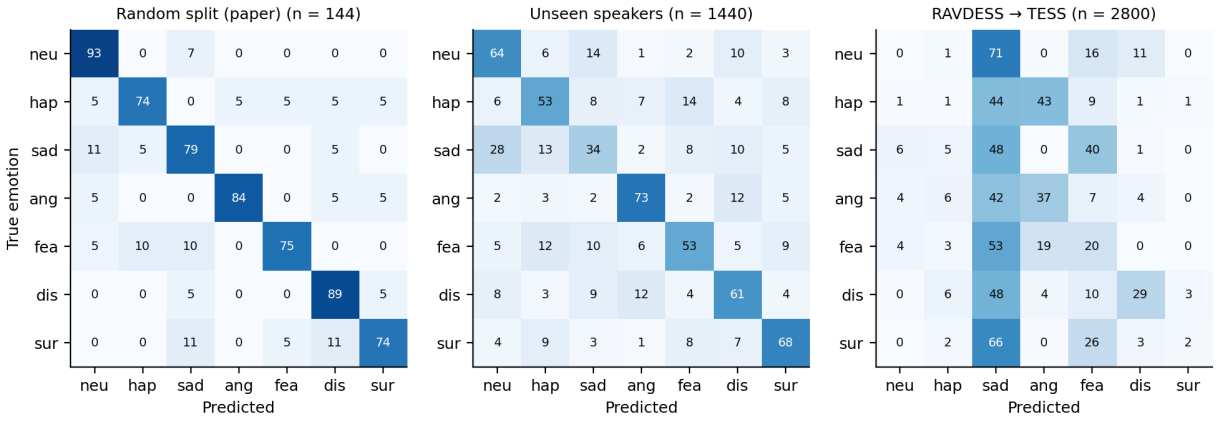


Figure 4. Ensemble confusion matrices (% of each true emotion): random split, unseen speakers, unseen dataset.

- **Easy and hard emotions.** For unseen speakers angry (73.4% recall), surprised (67.7%) and neutral (64.2%) are best; **sad is worst (34.4%)**. The most frequent errors are sad → neutral (27.6% of sad clips), neutral → sad, neutral → disgust, happy → fearful and angry → disgust: emotions with similar energy and pitch are confused.
- **Intensity:** strongly expressed emotions are recognised much better (66.1% vs 47.9%, $p < 0.001$).
- **Speakers:** per-actor accuracy ranges from 40.0% to 81.7% (Figure 5).
- **New dataset:** on TESS the ensemble predicts *sad* for 53.1% of clips and almost never neutral, happy or surprised. TESS clips are short, so after the 0.6 s offset 41.5% of their frames are zero padding (RAVDESS: almost none), which looks like quiet, sad speech. Loading TESS from 0 s instead raises the same ensemble from 19.7% to 30.5% and cuts sad predictions to 11.1%; the rest of the gap comes from different speakers, words and recording conditions. The older TESS speaker is recognised worse (15.5% vs 23.9%).

8. Bias and Fairness Check (Gender)

Unseen-speaker protocol, so every clip is judged by a model that never heard that actor; 720 clips per gender, so the data are balanced.

Model	Male actors	Female actors	Gap (F – M)	p-value
SVM	45.3%	52.9%	+7.6	0.004
Random Forest	41.0%	52.6%	+11.7	< 0.001
1D-CNN	49.4%	63.9%	+14.4	< 0.001
CNN-BiLSTM	47.2%	65.8%	+18.6	< 0.001
Ensemble	50.7%	66.2%	+15.6	< 0.001

Table 3. Accuracy by gender (two-proportion z-test).

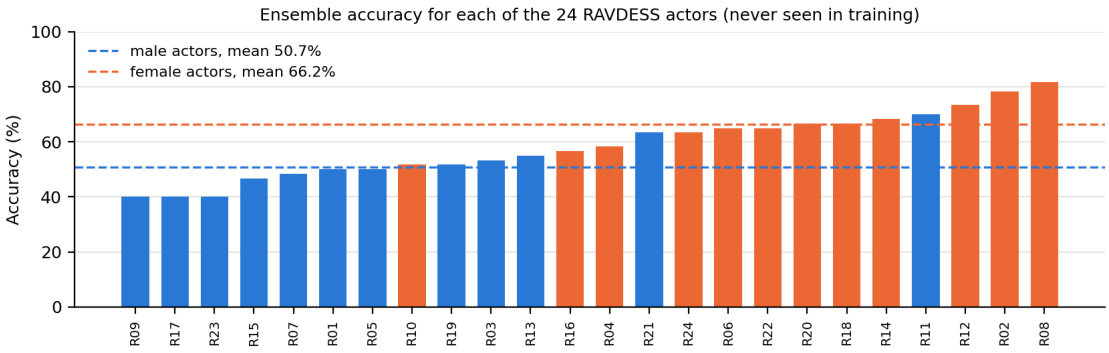


Figure 5. Ensemble accuracy per actor (blue = male, orange = female).

Every model is worse for male voices, and the deep models have the largest gaps (up to 18.6 points). The result holds at actor level (12 vs 12 per-actor accuracies, Mann–Whitney $U = 129$, $p = 0.001$; 10 of the 12 lowest actors are male). The gap is largest for surprised (+31.2), neutral (+20.1) and disgust (+19.8) and reversed only

for happy (−2.1). With a random split, where every actor is in training, the gap is small and not significant (80.0% vs 84.4%, p = 0.50): the bias appears exactly in the realistic case of a new user.

9. Feature and Augmentation Ablation

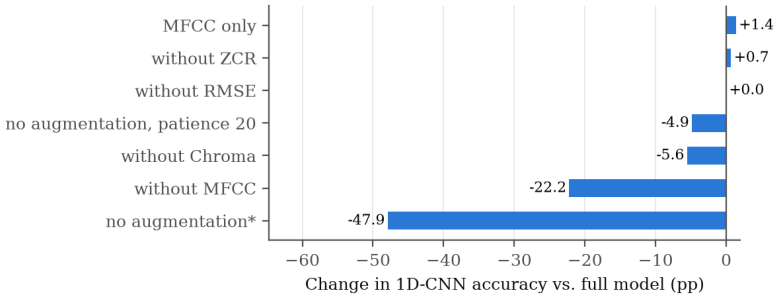


Figure 6. Change in 1D-CNN accuracy when parts of the input are removed (random split, 144 test clips; one standard error ≈ 3.4 points). *With the paper's early stopping, which halts before the model learns.

- **MFCC is essential:** without it the 1D-CNN falls from 79.2% to 56.9% and the SVM from 77.1% to 45.1%. MFCC alone (80.6%) is as good as all features.
- **ZCR and RMS add nothing measurable** (changes of +0.7 and 0.0 points).
- **Chroma is inconsistent:** removing it changes the 1D-CNN by -5.6 and the SVM by +2.1 points, within noise; the paper's claim that Chroma contributes substantially is not confirmed.
- **Augmentation and early stopping interact:** without augmentation the 1D-CNN stopped at epoch 9 with validation accuracy still at chance (31.2%), because an epoch has 4× fewer updates. With patience 20 it reaches 74.3%, so augmentation helps by only about 5 points.

10. Theoretical Insights (Follow-up Experiments)

To explain the results we ran four cheap experiments with the SVM on the same features (`phase5/F26-11.ipynb`).

- **The features describe the speaker more than the emotion.** On average the speaker explains 27.9% of a feature's variance and the emotion 7.6% (eta squared; speaker dominates in 65 of 68 features). The same SVM identifies **who is speaking with 95.0%** accuracy (chance 4.2%) but the emotion with 74.4%. A model can therefore score well on familiar speakers by learning each voice's typical values, which fail for a new voice.
- **Per-speaker normalisation helps new speakers and new datasets.** Z-scoring each speaker's features with that speaker's own statistics (no labels needed) raises the SVM from 49.3% to **59.2%** for unseen speakers (as good as the deep ensemble) and from 24.2% to **34.8%** on TESS. It does *not* close the gender gap (7.5 → 11.0 points), so the bias has another cause than different voice baselines.
- **The flattened input is tied to time positions.** Delaying the test speech by only 0.12 s costs an SVM on the paper's 3,672-value input 15.3 points but an SVM on the 68-value summary only 2.1. This explains why the summary-based Random Forest transfers best to TESS and why padding hurts the deep models.
- **A 144-clip test set is noisy.** Ten random splits give the same SVM 72.9%–83.3% (std 4.1 points; theory: 3.5). Differences of a few points between models or runs are within this noise, which is why we rely on confidence intervals and the 1,440-clip speaker protocol.

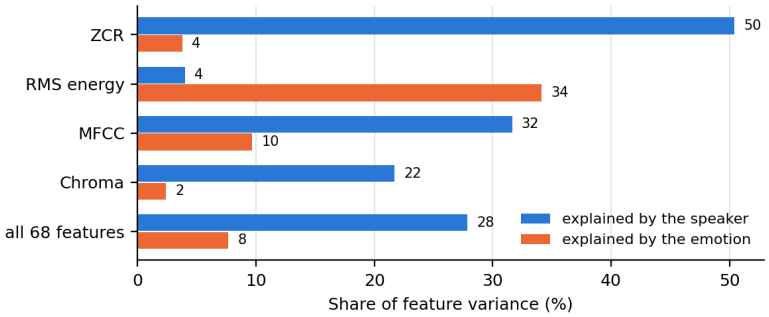


Figure 7. Share of feature variance explained by the speaker and by the emotion.

11. Working Product and Reproducibility

Demo app. `streamlit run app/app.py` opens a web page where the user uploads a .wav file; the app extracts the same 3,672 features, normalises them with the saved training statistics, runs both saved models, averages their probabilities and shows the predicted emotion with a bar chart of all 7 probabilities, plus a warning that accuracy is lower for other speakers and recording conditions. The features it computes from an uploaded file are identical to the cached training features (maximum difference 0.0). In Phase 3, the notebook's prediction function classified 5 of 6 raw test files correctly.

Reproducibility. All seeds are fixed (42) and every per-clip prediction is saved. All 30 reported accuracies are recomputed from these files, and re-training the SVM on another machine reproduces 100.0% of its random-split predictions and 98.7% of its unseen-speaker predictions. Deep models and the Random Forest vary by a few points with hardware and library versions (Phase 3 re-run: ensemble 77.1%, inside the 95% interval of 75.7–88.2%).

12. Challenges Faced

- **Environment setup.** TensorFlow 2.21 is about 1.5 GB; on our slow connection pip's dependency resolver appeared to hang until we pinned the exact version. Jupyter first started the system Python without TensorFlow, so we had to register a kernel for the correct environment. The demo app's temporary-file handling did not work on Windows and was changed to read the upload from memory.
- **Compute.** Convolving a 3,672-step input with 128 filters is slow without a GPU: about 3.7 min per epoch on our 2-core laptop, so a single training of both deep models takes about 2.8 hours. The 23 runs of the full study took roughly 4 hours on a 4-core machine; the ablation therefore uses one split, and the follow-up experiments use the fast SVM.
- **Training convergence.** Both deep models overfit: training accuracy reaches about 99% within 10 epochs while validation accuracy levels off at 70–80%. Dropout, batch normalisation, early stopping and learning-rate halving keep the best epoch. Without augmentation the paper's early-stopping patience stopped training before the model learned anything (31.2%); a longer patience fixed it.
- **Under-specified paper.** Split order, ensemble training and the label of RAVDESS *calm* are not stated, and the parameter table is inconsistent; we made each choice explicit and tested the split order.
- **Data issues.** Duplicate files in the Kaggle copies (removed by name), the unbalanced neutral class (hence macro-F1), and short TESS clips that become mostly zero padding.
- **Honest evaluation.** A 144-clip test set cannot separate models a few points apart, and clips of one actor are not independent; we added confidence intervals, McNemar and actor-level tests.

Limitations and future work. Both corpora contain acted North-American English speech, so even our strictest results overestimate performance on natural speech; TESS has only two (female) speakers, so the gender analysis uses RAVDESS only. Next steps are per-speaker normalisation inside the deep models, features that do not depend on time position, multi-corpus training and a larger, more diverse corpus such as CREMA-D (91 speakers).

13. Instructions to Run the Code and Reproduce the Results

Full instructions are in the repository README. In short (Python 3.13, CPU only):

Command	Purpose
`git clone https://github.com/junaidshahid-dev/Ai-project.git`	get the code
`pip install -r requirements.txt`	install the libraries (TensorFlow only needed for training / app)
`download RAVDESS and TESS from Kaggle into data/RAVDESS and data/TESS`	datasets (links in README)
`python scripts/extract_features.py`	build the feature cache (about 10 min)
`streamlit run app/app.py`	demo: upload a .wav file, see the predicted emotion
`python scripts/run_experiments.py paper leaky speaker cross ablation ablation_control`	all training runs and saved predictions (about 4 h on 4 cores)
`python scripts/analyze.py`	tables, figures and results/summary.json
`run phase4/F26-11.ipynb and phase5/F26-11.ipynb`	evaluation (under 1 min) and analysis (about 6 min)

14. Conclusion

We re-implemented a recent lightweight speech emotion recogniser and asked how far its near-perfect accuracy generalises. Under a leak-free split it reaches 81.9%, not 97.57%; the reported level is reproduced only when augmented copies of test clips are in training. For new speakers accuracy drops to 58.5%, across datasets it is near chance, and it is significantly worse for male speakers. The features carry more information about the speaker than the emotion; per-speaker normalisation is a simple remedy for new speakers and datasets, while the gender gap needs further work. For practitioners the lesson is concrete: split before augmenting, report speaker-independent and cross-corpus results with confidence intervals, and check accuracy per demographic group before deploying an emotion recogniser.

References

[1] J. H. Chowdhury, S. Ramanna, K. Kotecha, “Speech emotion recognition with light weight deep neural ensemble model using hand crafted features,” *Scientific Reports* 15, 11824 (2025), doi:10.1038/s41598-025-95734-z. [2] S. R. Livingstone, F. A. Russo, “The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS),” *PLoS ONE* 13(5), e0196391 (2018). [3] M. K. Pichora-Fuller, K. Dupuis, “Toronto Emotional Speech Set (TESS),” Borealis (2020). [4] B. McFee et al., “librosa: Audio and music signal analysis in Python,” *SciPy* (2015). [5] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *JMLR* 12 (2011). [6] M. Abadi et al., “TensorFlow: Large-scale machine learning on heterogeneous systems” (2015). [7] S. Kapoor, A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns* 4(9) (2023).